Network Science

Lecture 07: Small-World Networks

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Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

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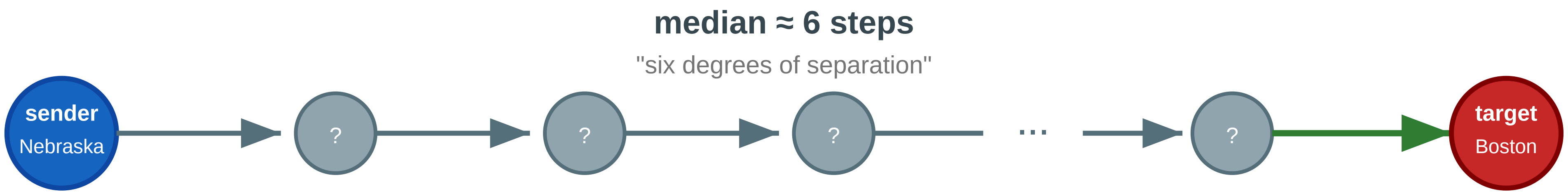
The letters that arrived took a **median of about 6 steps**.

The phrase "six degrees of separation" entered popular culture — but the experiment raised more questions than it answered.

The Chain

Milgram's small-world experiment (1967)

Can a letter reach a stranger in Boston through a chain of personal acquaintances?



Questions this experiment raises

① Why do short paths exist at all in a network of 200 million people?
② How can a network be locally clustered AND have short paths to strangers?
③ If short paths exist, can ordinary people find them using only local knowledge?
④ Only ~25% of chains arrived. Why did most fail?

> Each intermediary sees only their own contacts and must guess who is "closer" to the target. About 25% of chains completed; the rest died when someone refused to forward. The experiment demonstrates both that short paths *exist* and that finding them is *hard*.

Four Questions from One Experiment

1. **Existence.** Why do short paths exist at all in a network of 200 million people?
2. **Coexistence.** How can a network be *locally clustered* (your friends know each other) and *globally short* (six steps to a stranger in another state)?
3. **Navigability.** If short paths exist, can ordinary people find them using only local knowledge — no map, no search engine?
4. **Failure.** Only ~25% of chains arrived. Does this mean short paths are rare, or that *finding* them is the bottleneck?

Every concept in today's lecture answers one of these.

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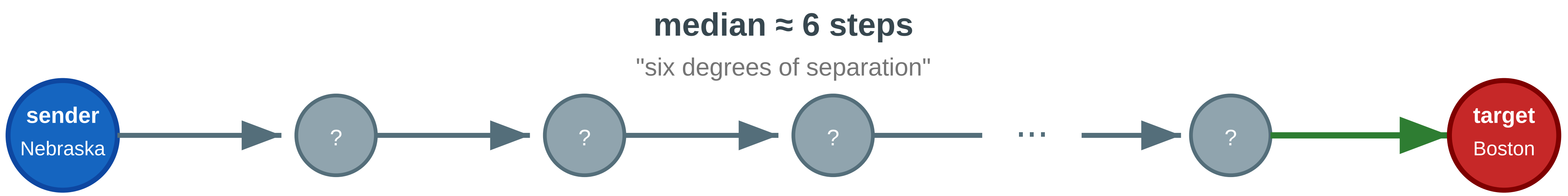

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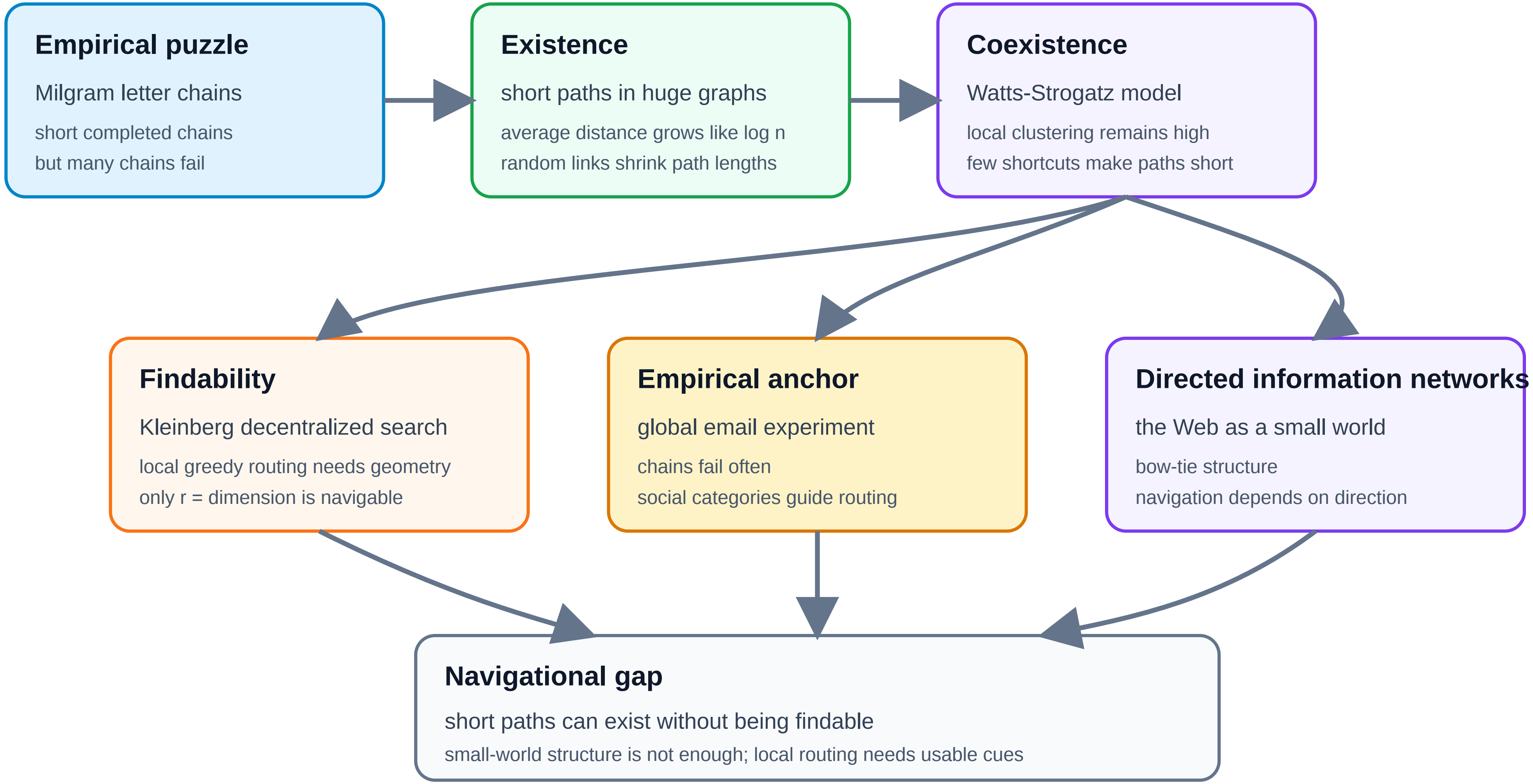
Argument Map



A New Kind of Gap — Existence vs. Findability

L03–L04 had a **computational** gap (NP-hard ideals). L05 had a **causal** gap. L06 had a **structural** gap (completeness required).

L07 has a **navigational** gap: short paths may *exist* without anyone being able to *find* them using local information.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance
L07	Navigational	Short navigable paths	Local information only	Geometric link distribution

Takeaway

> The small-world phenomenon is really two phenomena: the *existence* of short paths (explained by a few random long-range links) and the *findability* of those paths (explained only by a specific geometric distribution of links). Milgram's experiment revealed both — and the gap between them.

The Small-World Phenomenon

Guiding Question: Why are distances in large networks often surprisingly short?

Logarithmic Distances

Small-world property. A network exhibits the small-world property if the average shortest-path distance $\bar{d}$ grows at most logarithmically with the number of nodes:

$$\bar{d} \propto \log |V|$$

Recall: $d_G(u, v)$ is the **graph distance** between two nodes — the number of edges in a shortest path. The average shortest-path distance is the graph-level average over node pairs: $\bar{d} = \frac{1}{|V|(|V|-1)} \sum_{u \neq v} d_G(u, v)$

In a random graph $G(n, p)$ with average degree k, the typical distance is approximately $\bar{d} \approx \frac{\log n}{\log k}$.

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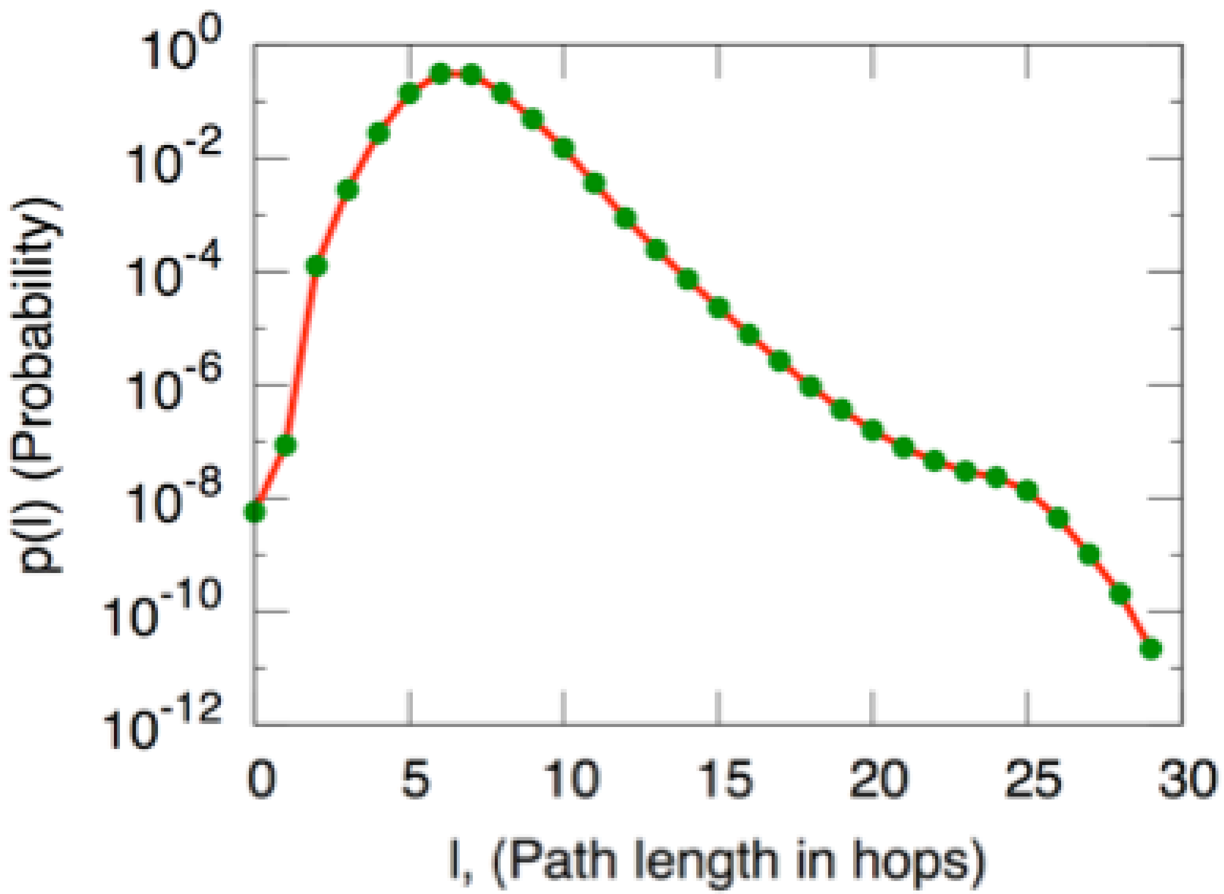
For a social network of $n = 300$ million people with average degree $k \approx 100$: $\bar{d} \approx \frac{\log(3 \times 10^8)}{\log 100} \approx \frac{19.5}{4.6} \approx 4.2$. Milgram's 6 is in the right ballpark.

Empirical Examples



Source: Easley & Kleinberg 2010, p. Ch. 20

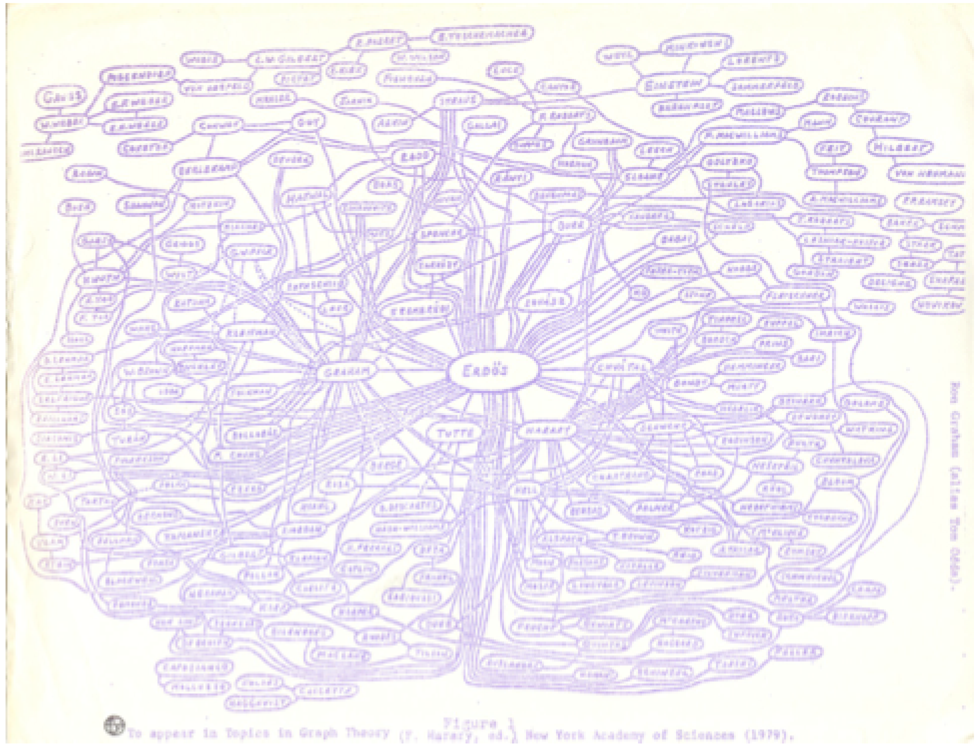


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdös [189]. (Image from http://www.oakland.edu/enp/cgraph.jpg)

Source: Easley & Kleinberg 2010

Left: Microsoft Instant Messenger network — 180 million users, median distance ~7. **Right:** Mathematical collaboration network — Erdős numbers rarely exceed 5. Both confirm logarithmic scaling across orders of

Takeaway

> Logarithmic distances are a robust empirical fact across social, communication, and collaboration networks. In any network with moderate average degree, even billions of nodes remain only a handful of hops apart.

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Quick Check

The Facebook social graph has approximately 3 billion users and an average degree of about 300.

1. Estimate the average shortest-path distance using $\bar{d} \approx \log n / \log k$.
2. Facebook reported an empirical average distance of 3.57 in 2016. Is this consistent with your estimate?
3. Why might the empirical value be *lower* than the logarithmic estimate?

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Quick Check — Solution

1. $\bar{d} \approx \frac{\log(3 \times 10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx 3.8$

2. Yes — 3.57 is remarkably close to the logarithmic estimate.

3. The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

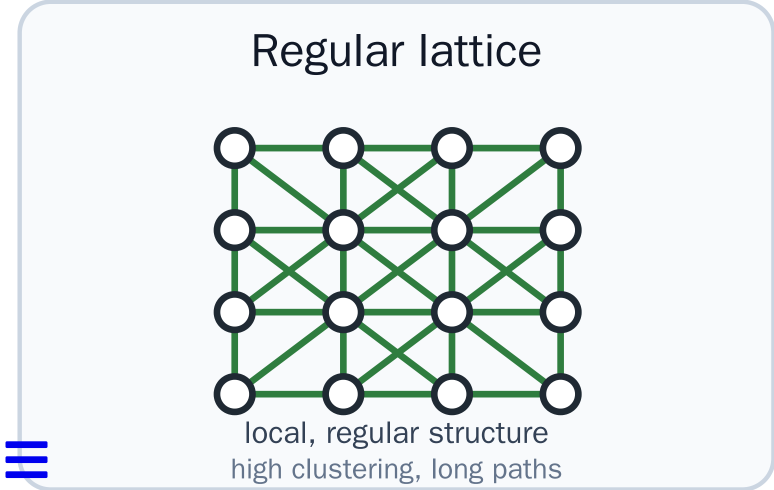
The Watts–Strogatz Model

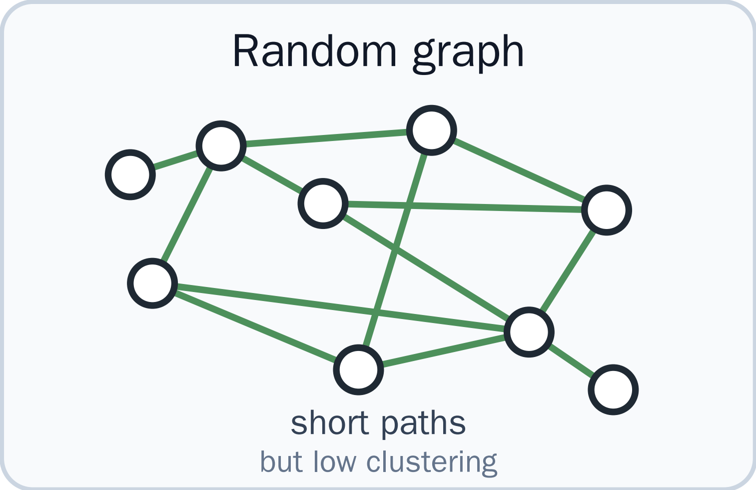
Guiding Question: How can a network be both highly clustered *and* have short paths?

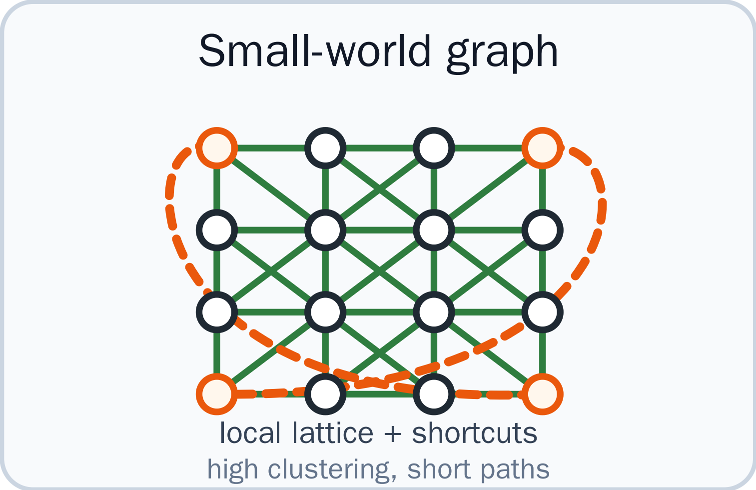
The Paradox

Real social networks have two seemingly contradictory properties:

Property	Regular lattice	Random graph	Real social network
Clustering C	High (~ 0.5)	Low ($\sim k/n$)	**High** (~ 0.1–0.5)
Avg. path length L	Long ($\sim n/2k$)	Short ($\sim \log n/\log k$)	**Short** ($\sim \log n$)







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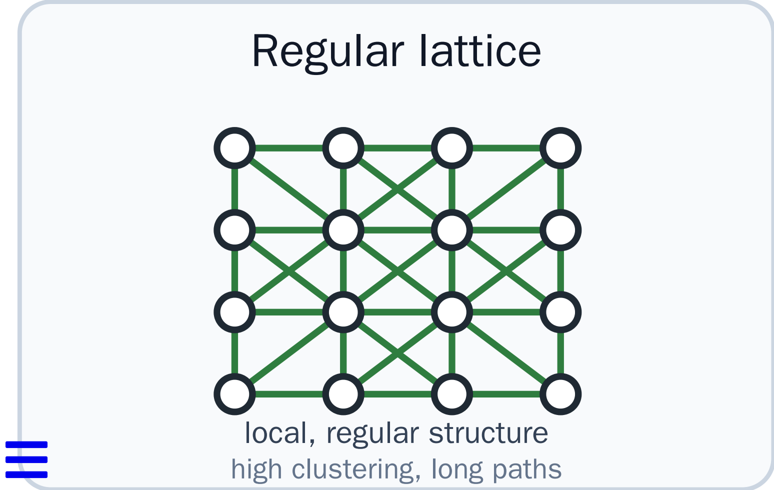
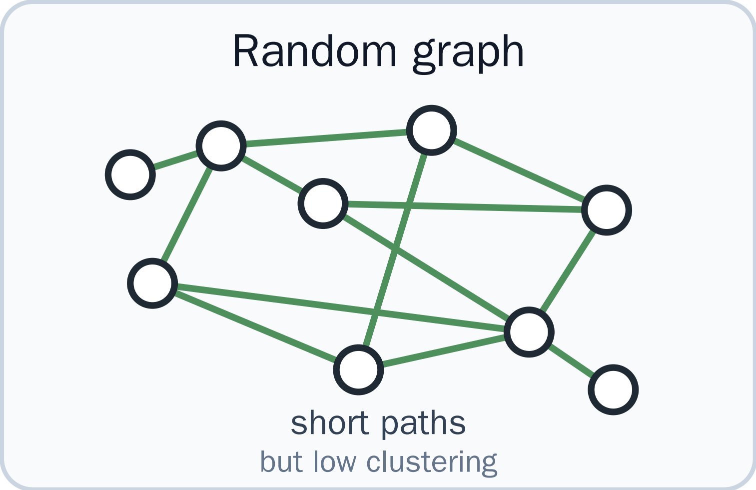
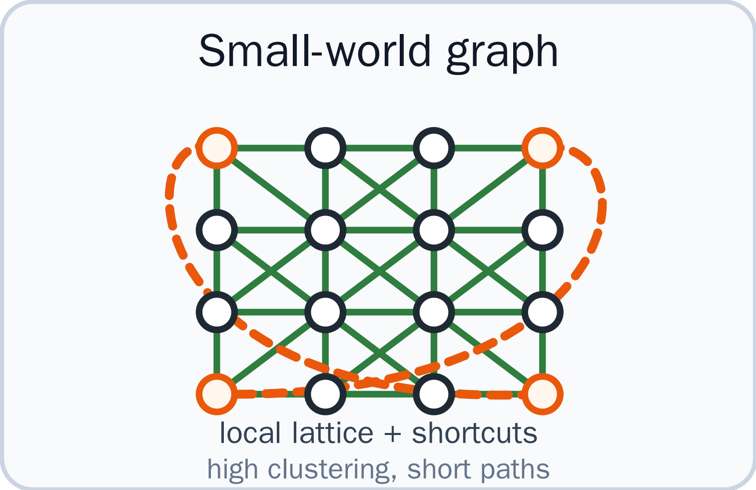
A regular lattice is clustered but large. A random graph is small but unclustered. **Real networks are both** — high C and low L simultaneously.







The Model

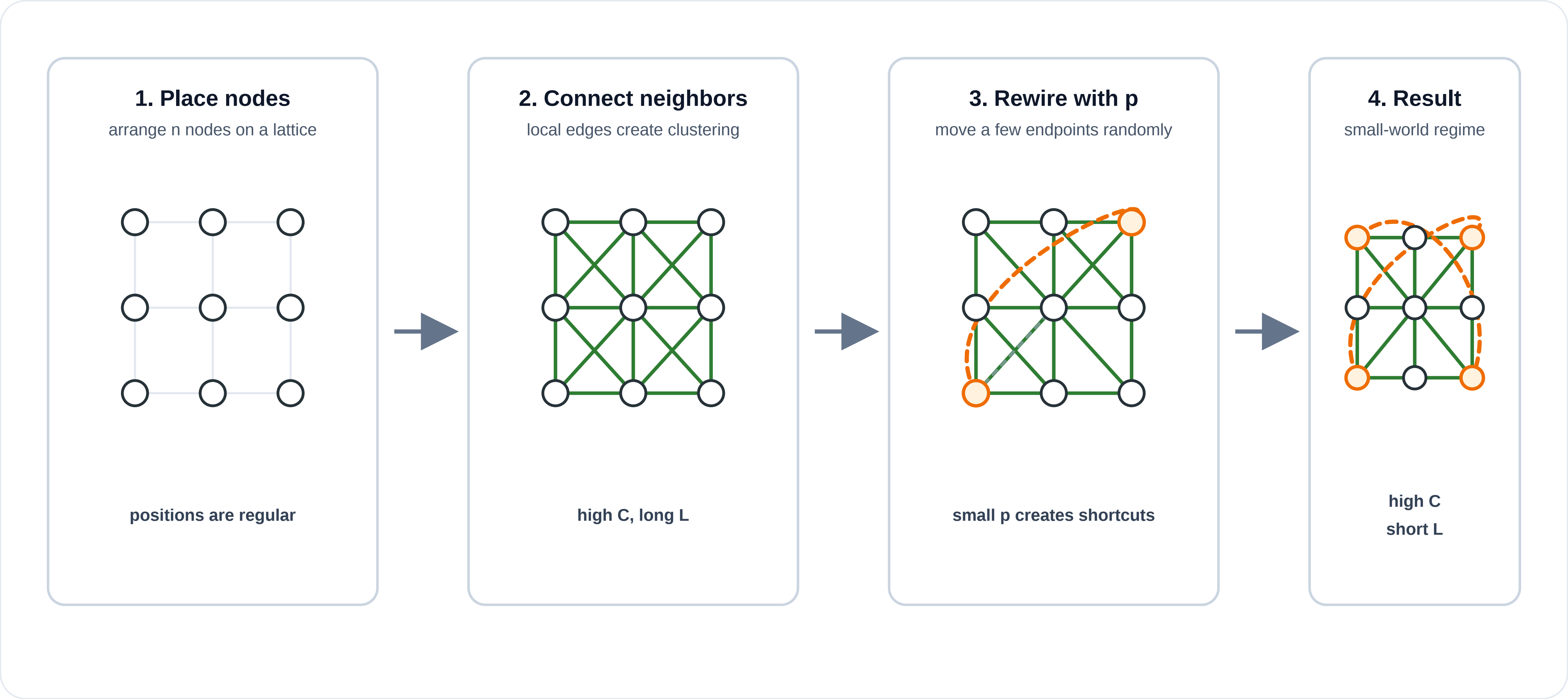
Watts–Strogatz model (1998).

1. Start with a **regular lattice**: n nodes arranged in a regular local structure, each connected to nearby neighbors.
2. For each edge, with probability p, **rewire** one endpoint to a uniformly random node (avoiding self-loops and duplicates).

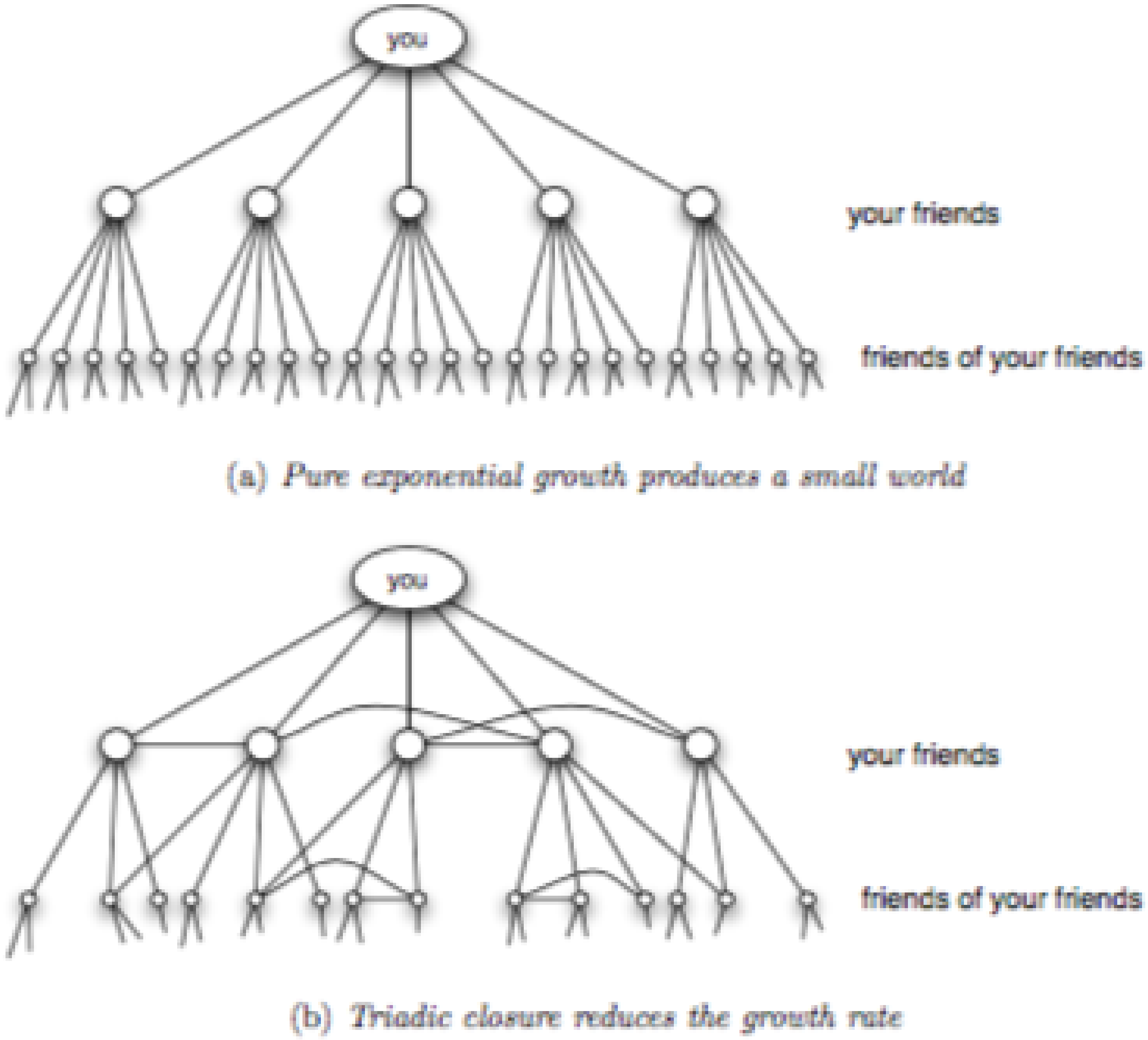
The parameter $p \in [0, 1]$ controls the interpolation:

- $p = 0$: regular lattice (high C, high L)
- $p = 1$: random graph (low C, low L)
- $0 < p \ll 1$: **small-world regime** (high C, low L)

Watts–Strogatz in Four Steps



Why Clustering Makes Paths Longer



Source: Easley & Kleinberg 2010, p. Ch. 20

If every friend led to entirely new people, reachability would grow like a tree and paths would be very short. In real friendship networks, friends of friends often overlap, creating triangles and high clustering. That overlap slows the expansion of the search frontier.

Takeaway

The Watts-Strogatz model resolves the clustering/path-length paradox: a tiny fraction of random long-range shortcuts collapses distances without destroying local structure. The small-world regime exists because $L(p)$ falls much faster than $C(p)$.

Quick Check

A regular local lattice has $n = 1000$ nodes and $k = 10$ neighbors per node.

1. At $p = 0$, should the average path length L be closer to a lattice scale or a random-graph scale?
2. After rewiring 1% of edges ($p = 0.01$), how many long-range shortcuts are created?
3. Qualitatively, what happens to L and C?

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Quick Check — Solution

1. Closer to a **lattice scale**: paths are still local and grow with the lattice diameter, not with $\log n$.

2. $nk/2 = 5000$ edges total; 1% = **50 long-range shortcuts**.

3. L drops sharply because shortcuts jump across the lattice. C barely changes because only 50 of 5000 edges were rewired, so the vast majority of local neighborhoods survive. **This is the small-world regime.**

Kleinberg's Decentralized Search

> **Guiding Question:** Short paths exist — but can local actors *find* them without a map?

The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

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Watts-Strogatz does not guarantee this works. Short paths exist, but the greedy algorithm may not find them — it depends on *where* the long-range links point.

Kleinberg's Grid Model (2000)

Setup. Place n nodes on a d-dimensional grid. Each node has:

- **local contacts:** all grid neighbors within lattice distance p *(the local radius; in 2D, usually Manhattan distance $|x_u - x_v| + |y_u - y_v|$)*
- **one long-range link:** drawn with probability proportional to $d(v, u)^{-r}$, where d is grid distance and $r \geq 0$ is the distance exponent.

Greedy routing. At each step, forward to the neighbor (local or long-range) closest to the target in grid distance.

The parameter r controls the *geometry* of long-range links:

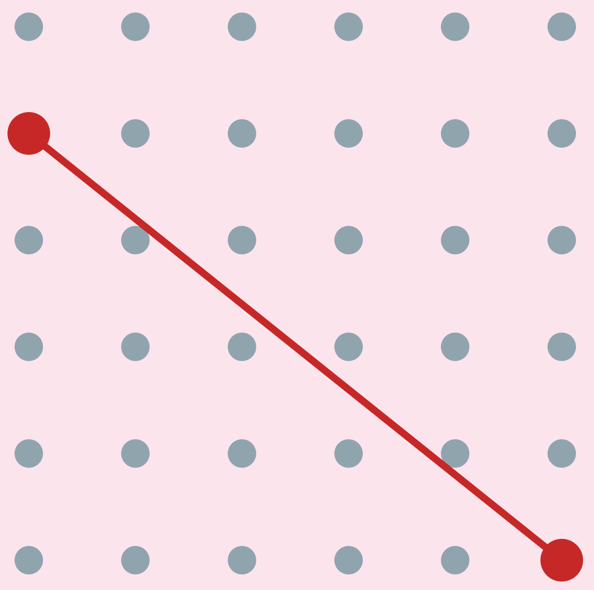
- $r = 0$: uniform random (long-range links go anywhere)
- $r = d$: inverse-power law matched to dimension (links span all scales equally)
- $r \gg d$: long-range links are effectively local

The Three Regimes

Kleinberg's grid model — the geometry of navigability

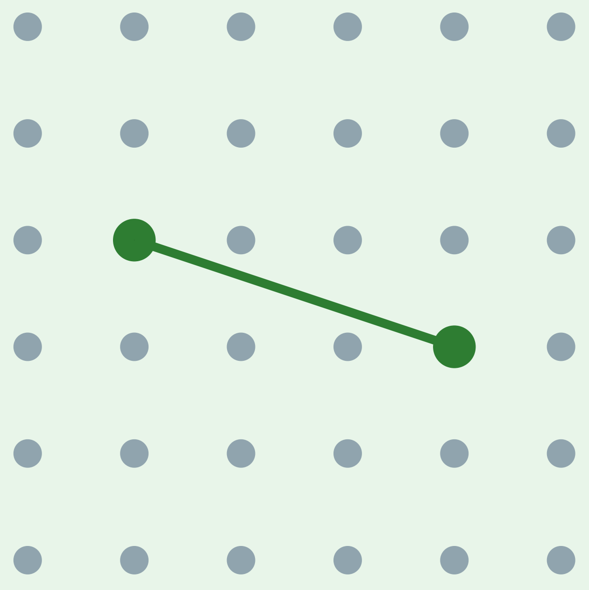
Each node has local grid neighbors + one long-range link drawn with $P(\text{link to } u) \propto d(v,u)^{-r}$

$r = 0$ (uniform random)



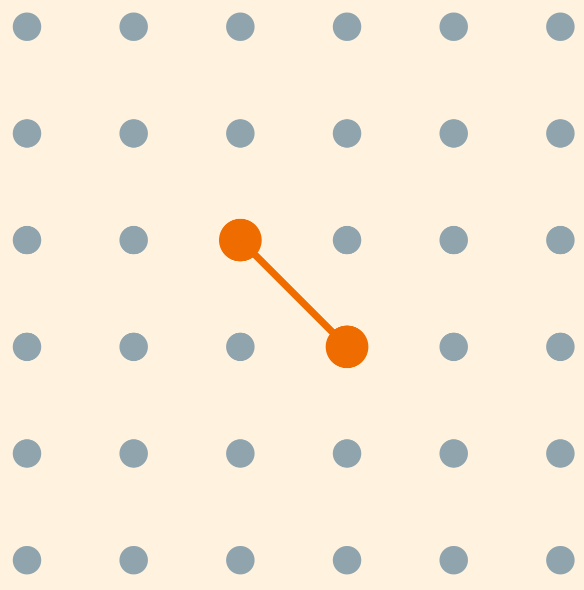
short paths exist,
but search is $\Omega(n^{2/3})$

$r = 2 = d$ (inverse-square)



short paths exist
AND search is $O(\log^2 n)$

$r \gg d$ (too clustered)



short paths exist,
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Kleinberg's Theorem

Theorem (Kleinberg, 2000).

In the d-dimensional grid model with long-range exponent r:

- If $r = d$: greedy routing finds the target in $O(\log^2 n)$ steps.
- If $r \neq d$: decentralized search has a **polynomial lower bound**: at least $\Omega(n^c)$ steps for some fixed constant $c = c(r, d) > 0$.

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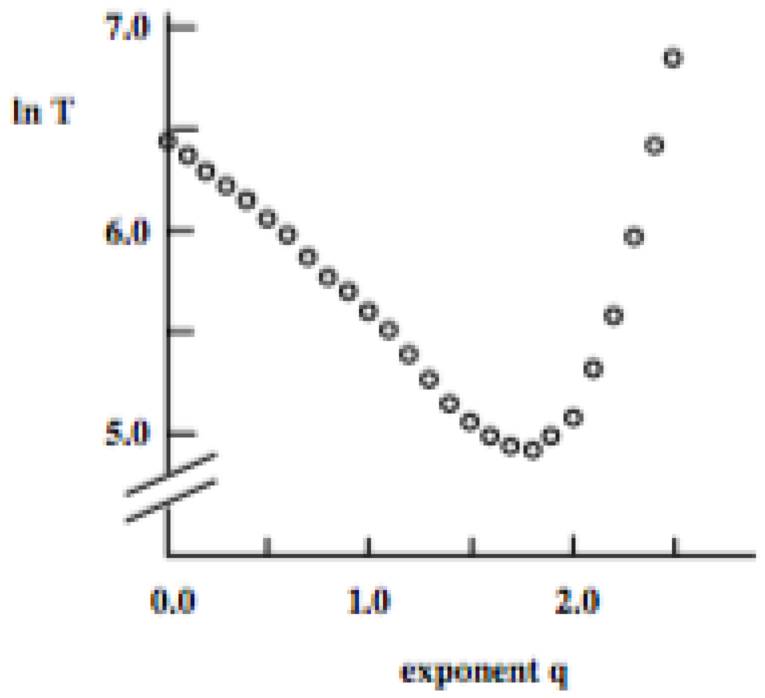
The exponent $r = d$ is **unique** — it is the only value at which decentralized search is polylogarithmic. Any fixed positive power n^c eventually grows faster than $\log^2 n$, even if c is small.

Simulation Evidence



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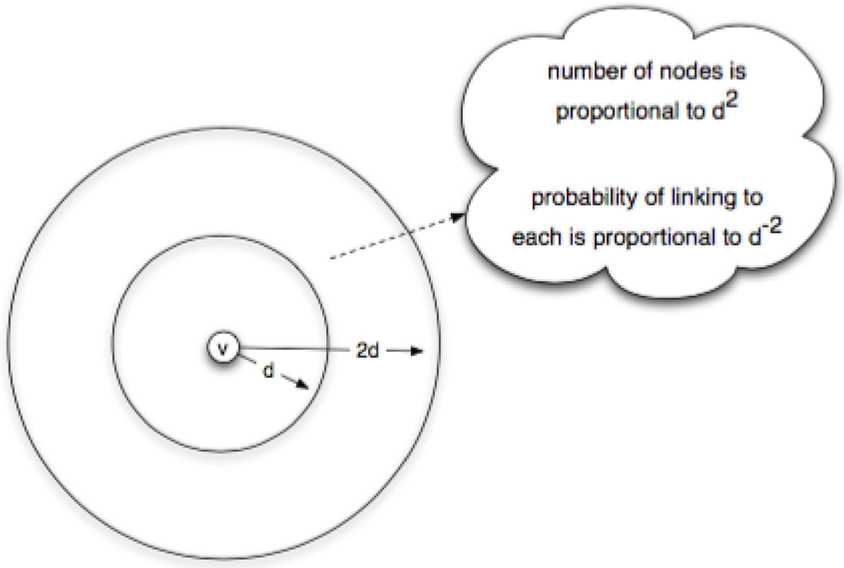

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> **Left:** delivery time vs. exponent r — a sharp minimum at $r = d = 2$. **Right:** the proof intuition — at $r = d$, each "ring" of nodes at distance 2^i from the target receives roughly equal link probability, creating a hierarchy of scales the algorithm descends one level at a time.

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Takeaway

Short paths exist in almost any small-world network. But navigability — the ability of local actors to *find* those paths — requires a specific geometric distribution of long-range links ($r = d$). Watts-Strogatz gives existence; Kleinberg gives findability.

Quick Check

A social network is well-modeled by a 2D grid with random long-range links.

1. What exponent r makes greedy search efficient?
2. If the network's long-range links are uniformly random ($r = 0$), what happens to (a) the existence of short paths and (b) greedy routing performance?
3. Why is the Watts-Strogatz model insufficient to explain Milgram's result?

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Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths **still exist** — uniform random shortcuts reduce average path length.
- (b) Greedy routing **fails** — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not *navigable* short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

Kleinberg: nodes live on a known grid. Greedy routing works when links exist at multiple scales.
Vector search: documents or images live in an embedding space. The "distance" is learned from data, not given by a map.

HNSW intuition: build a graph over vectors with two kinds of movement:

- sparse upper layers for **large jumps**
- dense lower layers for **local refinement**

Search starts coarse, then descends and refines greedily near the query.

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> HNSW is not literally Kleinberg's grid. The useful analogy is the **multi-scale search structure**: jump far first, then use local links to finish.

Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into $\mathbb{R}^d$ (typically $d = 384$–4096).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
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> We will come back to **how** these embeddings are constructed in **Lecture 09** (node and document representations, spectral methods, DeepWalk, node2vec, GNNs). The *geometry* of the embedding space determines whether navigation works — which closes the loop to Kleinberg.

Empirical Anchor: The Global Email Experiment

> **Guiding Question:** Does Milgram's result replicate at scale — and what does chain failure tell us?

The Study

Dodds, Muhamad & Watts (2003), *An Experimental Study of Search in Global Social Networks*, [Science 301, 827–829].

They replicated Milgram's experiment using **email** — a global medium with no geographic constraint.

Parameter	Value
Participants recruited	**~60 000** from 166 countries
Target persons	**18** in 13 countries
Chains started	**24 163**
Chains completed	**384** (~1.6%)
Median chain length (completed)	**4–7 steps**

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What They Found

1. Completed chains are short. Median 4–7 steps, consistent with Milgram and with $\log n / \log k$ scaling.

2. Most chains die early. The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.

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3. Successful chains use geography and profession as search dimensions. Forwarders who brought the letter closer in *geographic* or *professional* distance were more effective — consistent with Kleinberg's model where search exploits multiple distance dimensions simultaneously.

What the Study Could Not See

1. Chain death ≠ path absence. A chain that dies after 3 steps does not mean no short path existed — it means someone chose not to forward. The study measures *willingness to search*, not *path existence*. The true completion rate under full cooperation is unknown.

2. Selection bias in completions. The 384 completed chains are a biased sample — they may systematically over-represent easy targets (well-known, geographically close, professionally distinctive). The median chain length of 4–7 steps may underestimate the true difficulty of reaching a random target.

3. Email ≠ acquaintance. Milgram's experiment required *personal acquaintance*; the email study allowed forwarding to anyone whose email address you knew. The social network traversed by email may be larger and weaker-tied than the face-to-face network Milgram assumed.

4. No ground-truth network. Without observing the full network, the study cannot measure whether the successful chains actually followed near-optimal paths or just happened to complete.

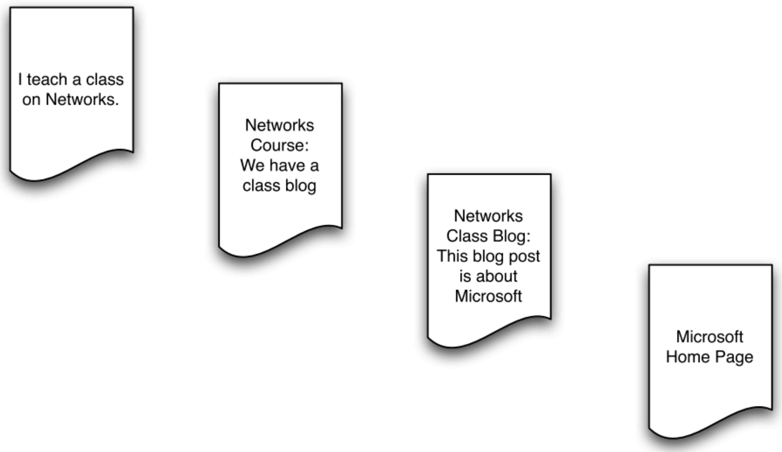
Takeaway

Dodds et al. (2003) confirmed Milgram at scale: short paths exist and people can sometimes find them. But the 1.6% completion rate reveals that *motivation*, not topology, is the primary bottleneck. The navigational gap is real — short paths exist but are rarely found in practice.

The Web as a Directed Small World

> **Guiding Question:** What happens to navigability when links are directed?

From Social to Information Networks



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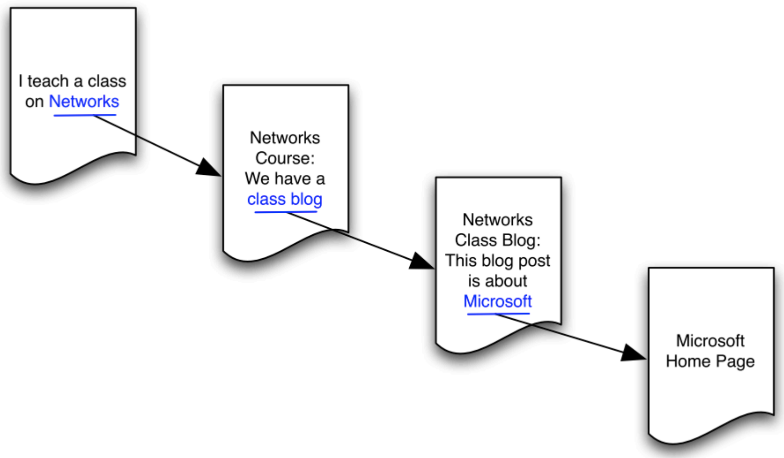

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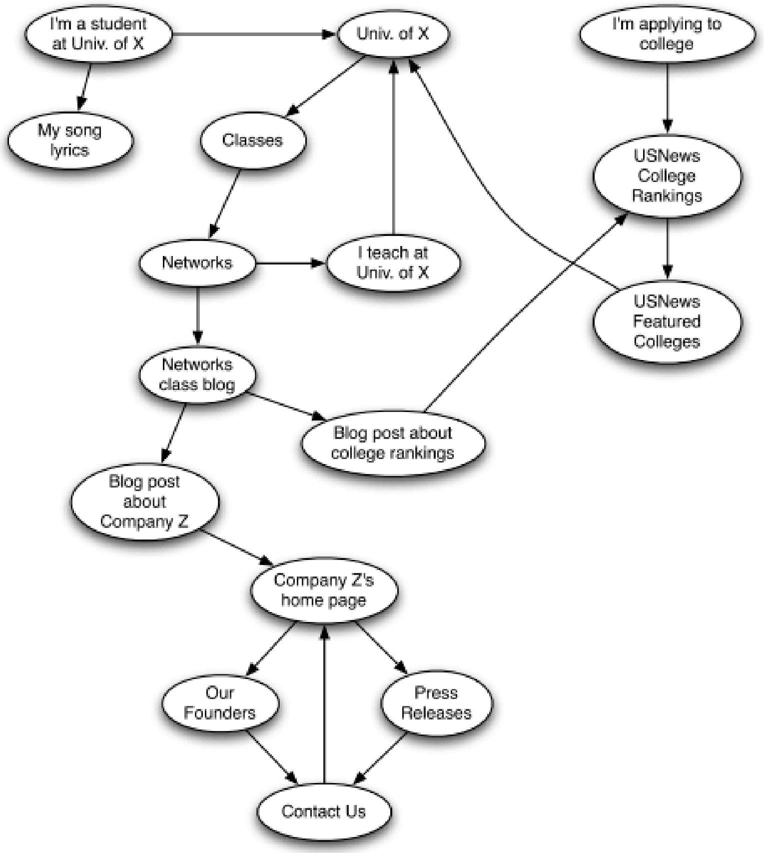
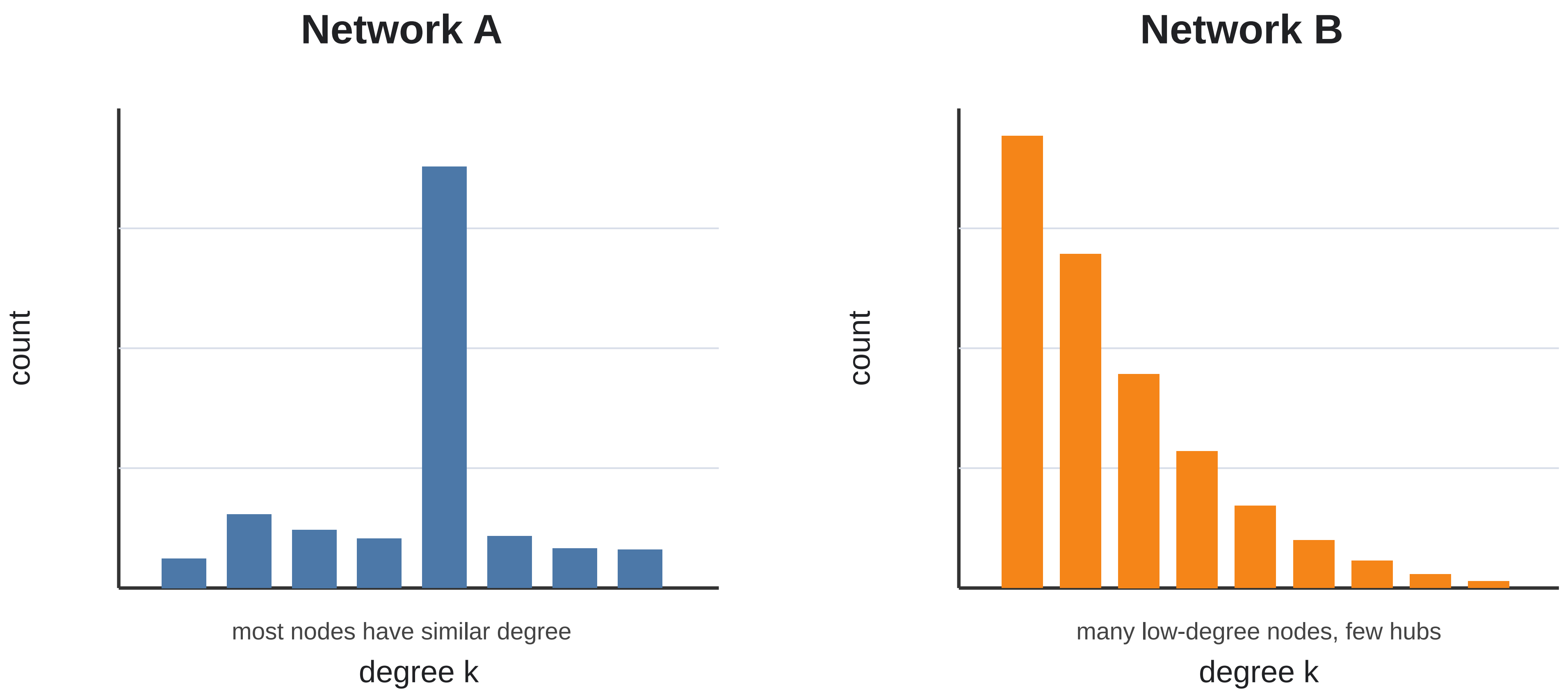


Source: Easley & Kleinberg 2010

> Hyperlinks are **directed** — page A can link to page B without B linking back. This means reachability is asymmetric: you can follow links *from* A *to* B but not necessarily in reverse.

Scale-Free Structure of the Web

The Web is not degree-homogeneous. Most pages receive few incoming links, while a small number of pages receive enormous numbers of links.



Scale-free-like Web. A directed information network whose in-degree distribution is heavy-tailed / approximately power-law:

$$P(k_{in}) \propto k_{in}^{-\alpha}$$

There is no single "typical" page: attention concentrates around hubs (Barab'asi & Albert 1999).

Barabási-Albert: A Model of Web Growth

Preferential attachment. New pages tend to link to pages that are already visible.

In a simplified directed Barabási-Albert model:

1. Start with a small connected Web.
2. Add one new page at a time.
3. Each new page creates m outgoing hyperlinks.
4. Existing page v receives a new link with probability proportional to its current in-degree:

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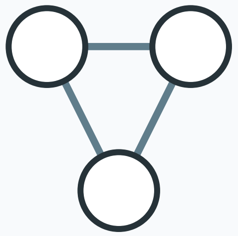
This is the **rich-get-richer** mechanism: popular pages attract more links, which makes them even more visible.

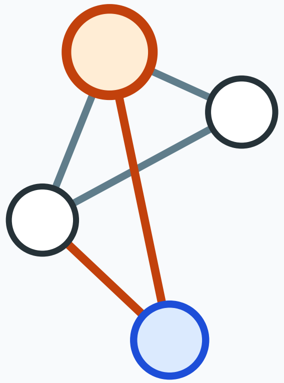
Barabási-Albert in Four Steps

1. Seed



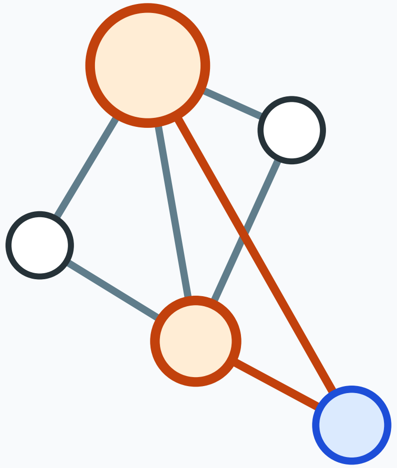
start with a small
connected graph

2. Grow



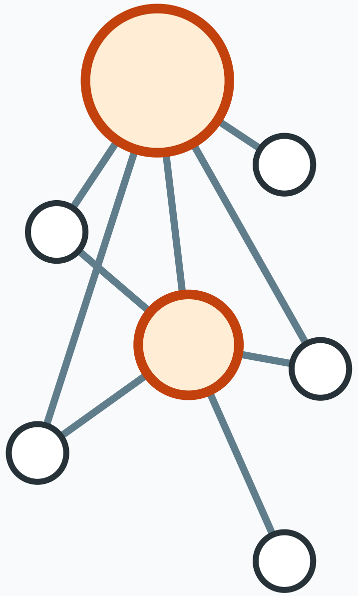
add one node
and m edges

3. Prefer hubs



choose endpoints with
probability proportional to degree

4. Hubs emerge



rich-get-richer growth
creates a heavy tail

Preferential attachment rule: probability of attaching to node v is proportional to its current degree k_v.

Nodes that are already well connected become even more likely to receive new edges.

What Preferential Attachment Explains

Web pattern	Preferential-attachment explanation	Limitation
Heavy-tailed in-degree	Popular pages attract more links	Not every popular page is high quality
Short paths to authorities	Hubs connect many topics	Direction still restricts reachability
Attention inequality	Visibility compounds over time	Aging and novelty also matter
Link copying	Authors imitate existing link lists	Communities and spam distort the signal

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> BA explains why hubs can emerge, but it is not a full model of the Web. The real Web also depends on content quality, topical communities, search engines, aging, copied links, platforms, and incentives.

Kleinberg on the Web: Structured Shortcuts

Scale-free hubs help explain **which pages become highly linked**. Kleinberg-style models explain a different question: **when can local navigation make progress toward a target?**

Model idea	Web interpretation
Preferential attachment	popular pages attract more incoming links
Kleinberg long-range links	hyperlinks connect across topical or semantic distance
Greedy routing	users follow links that seem closer to the information goal

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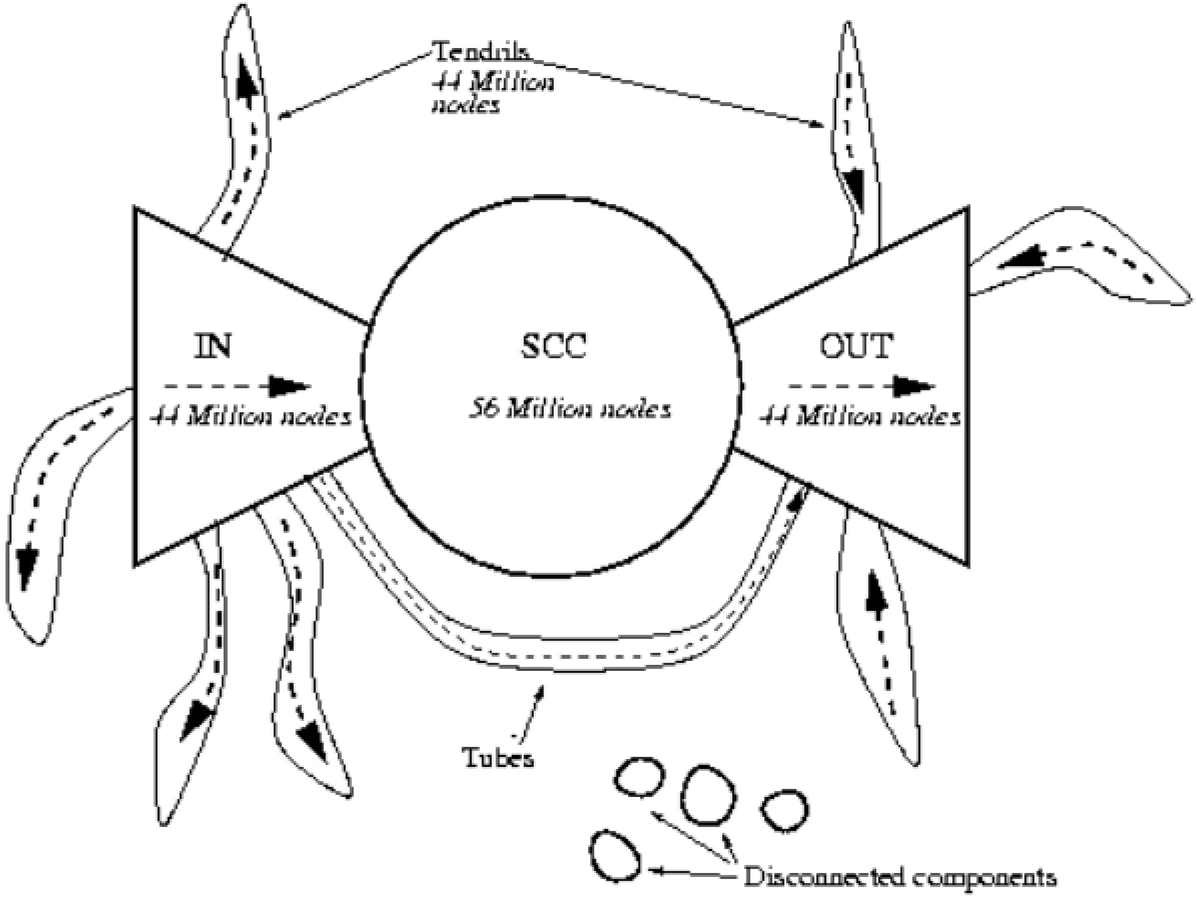
> Kleinberg does **not** imply scale-free. It uses a power law over **distance**, not over **degree**. The Web can be scale-free-like and small-world-like at the same time, but those are separate properties.

Web Synthesis

The Web can be seen as both **scale-free-like** and **small-world-like**. Preferential attachment explains the emergence of hubs, because popular pages attract more links over time. Kleinberg-style models explain navigability through structured shortcuts, especially when links reflect topical or semantic distance. However, the real Web is shaped by additional mechanisms such as content quality, topic communities, search engines, aging, and copied links.

The Bow-Tie Structure



Source: Easley & Kleinberg 2010, p. Ch. 13

Broder et al. (2000) crawled ~200 million web pages and found the Web decomposes into:

- **SCC (Strongly Connected Component):** ~28% of pages — every page reachable from every other via directed paths. *This* is the "small world" of the Web.
- **IN:** pages that can reach the SCC but are not reachable from it.
- **OUT:** pages reachable from the SCC but that cannot reach back.
- **Tendrils and tubes:** pages connected to IN or OUT but not to the SCC.

Navigability in Directed Networks

The small-world property and navigability both change in directed networks:

- **Short paths exist** within the SCC — the strongly connected core is a small world (~16 clicks between random page pairs within SCC).
- **Short paths do not exist** from OUT back to SCC, or between tendrils.
- **Greedy search is harder** because you can only follow outgoing links — you cannot "pull" information from a page that doesn't link to you.

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Search engines (Google, 1998) solve the navigability problem by building a *global index* — they observe the entire graph and compute optimal paths. This is the opposite of Kleinberg's decentralized search: it works precisely because it is *centralized*.

Takeaway

The Web's directed links create asymmetric reachability. Only the Strongly Connected Component (~28%) is a "small world." Search engines solve the navigability problem through centralized indexing — the opposite of Milgram-style decentralized routing.

Quick Check

A directed network has 1000 nodes. The SCC contains 300 nodes. The IN component has 200 nodes, and the OUT component has 250 nodes. The rest are tendrils.

1. From a random node in IN, can you reach a random node in OUT? Explain your path.
2. From a random node in OUT, can you reach a node in IN?
3. What fraction of all possible source-target pairs have a directed path between them?

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Quick Check — Solution

1. Yes: IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

2. No. OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are *not* mutually reachable.

Wrap-Up — Lecture 07

Question	Key idea	Takeaway
Why are large networks close?	Logarithmic distances	Short paths can exist even in huge networks.
How can networks be clustered *and* short?	Watts-Strogatz shortcuts	A few long-range edges collapse L while C stays high.
Can local actors find short paths?	Kleinberg's theorem	Findability needs the right multi-scale link geometry.
Do people actually complete searches?	Milgram; Dodds et al.	Many paths die because motivation and information are limited.
What changes on the Web?	Directed bow-tie structure	Reachability is asymmetric; only the SCC behaves like a small world.
Why is the Web hub-dominated?	Barabási-Albert	Preferential attachment can produce scale-free-like in-degree.
Is the Web locally navigable?	Structured shortcuts	Topical/semantic links help, but search engines centralize navigation.

> Lecture 07 separates **existence** from **findability**: short paths may be present in the graph, but local actors can use them only when the network provides navigable structure.

Looking Ahead: Lecture 08

How do things *spread* on networks — diseases, information, behaviors? The structure we have built (ties, communities, balance, small-world shortcuts) now becomes the substrate for dynamic processes: contagion, diffusion, and cascading failure.

Reading

Chapter 20 of Easley & Kleinberg (2010) covers the small-world phenomenon and Watts-Strogatz. Barabási & Albert (1999) introduces scale-free networks and power-law degree distributions. Kleinberg (2000), *The Small-World Phenomenon: An Algorithmic Perspective*, is the source for the decentralized search theorem. Dodds, Muhamad & Watts (2003) is the email experiment. Broder et al. (2000) is the bow-tie structure of the Web.

Image Credits & References

Barab'asi, Albert-L'aszl'o and Albert, R'eka (1999). Emergence of Scaling in Random Networks. *Science*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/